

Cytology

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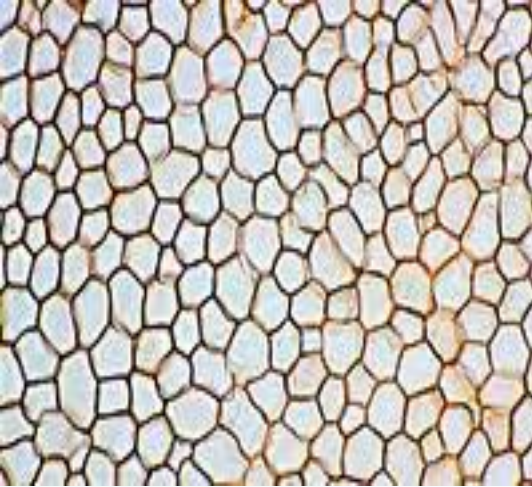
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Lecture.1

The cell definition and types

- The **cell** is the smallest structural and functional unit of the all living organisms.
- **Cell biology** is the study of the cell.
- **Cell biology** is also known as **Cytology**.
- The word cytology was derived from Greek word *Kitos* (cytos) means a compartment and *Logus* means to discover or to study.

Therefore **cytology** literally means a branch of biology which studies the different aspects of cells, like study of phenomenon of **heredity, variation, evolution, nutrition, metabolism, growth, reproduction** of cell, etc.



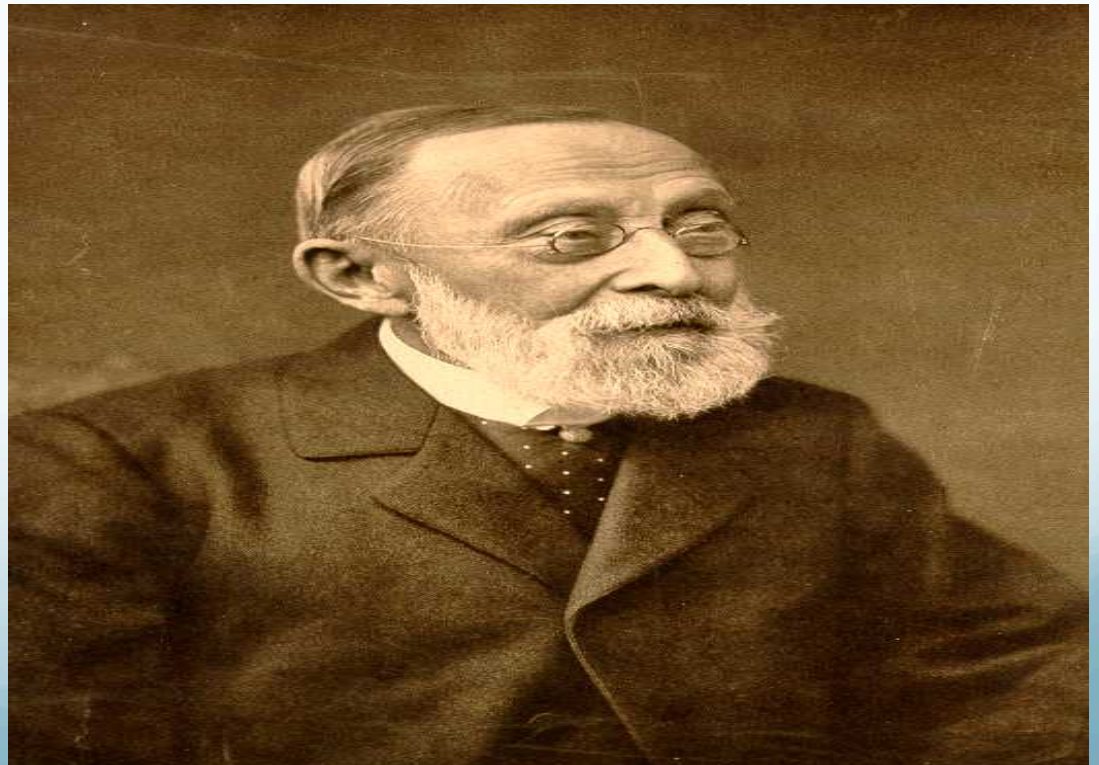
- **History**

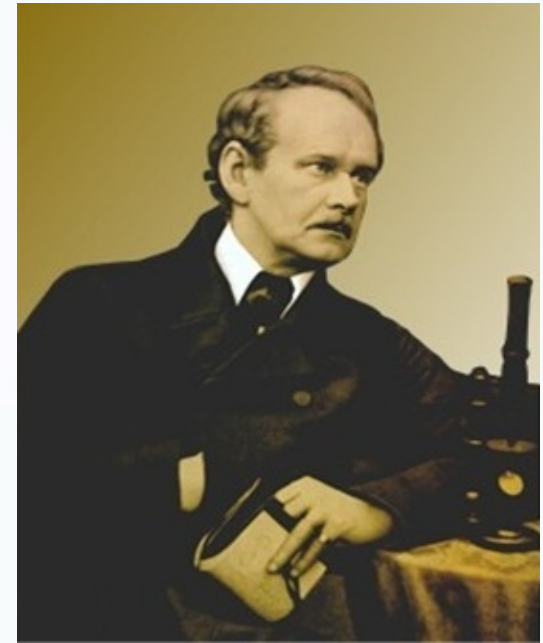
- 1. In **1665**, **Robert Hooke** first time observed the cell. He observed a thin slice of cork under his own microscope. Under microscope, he observed a honey-comb like structure showing hallow empty spaces or compartments surrounded by firm cell wall. To each hallow space he called a '*cell*'.
- Actually what Robert Hooke observed was dead cell.

2. *Antony Van Leeuwenhoek* (1632–1723) observed first time the microorganisms (prokaryotic cell) under his compound microscope. He observed tiny microbes in a drop of pond water which he called animacules.



3. In **1831**, **Robert Brown**, first time observed a nucleus in the cell.
4. In **1855** **Rudolph Virchow** observed that new cells arise from preexisting cells.





5. In **1838**, M.J. **Schleiden**, a German **botanist** studied many plants section under microscope and came to conclusion that, “all plants are ultimately made up of cell”.



6. In **1839** T.S. **Schwann**, a German **Zoologist** came to the similar conclusion that, “all the animals are ultimately made up of cell”.

Later, Schleiden and Schwann both together put forth a theory which is known as a **cell theory**. According to this theory all the organisms i.e. plants and animals are made up of cells.

The **cell theory** may be summarized as

- *(a)* All Organisms are composed of one or more cells and these cells are differentiated into distinct cell types.
- *(b)* A cell is a feature of all organisms with the exception of viruses (**cell is the basic unit of structure and organization of organisms**).
- *(c)* The protoplasm is the living content of the cell and determines the activity of the cell and thus the whole organism.

(d) New cells originate from preexisting cells through division.

- **All living cells/organisms have a common ancestor.**
- **Therefore, all cells shows common simplicity in the molecular organization and principle of molecular economy.**

In twentieth century, after the invention of modern microscopes and various technique, our knowledge of cell increased and a statement of cell theory was slightly modified.

According to the latest cell theory, “**all the living organisms consists of cells and cell products**”.

Properties of Cells

1- Cells are complex and highly organized

- They contain numerous internal structures
- Some are membrane bound (organelles) while others do not

2-Cells contain a genetic blueprint and machinery to use it

- Genes are instructions for cells to create specific proteins
- All cells use the same types of information
- The genetic code is universal
- Information transfer must be error free

Errors are called mutations

Genetic Code- Table

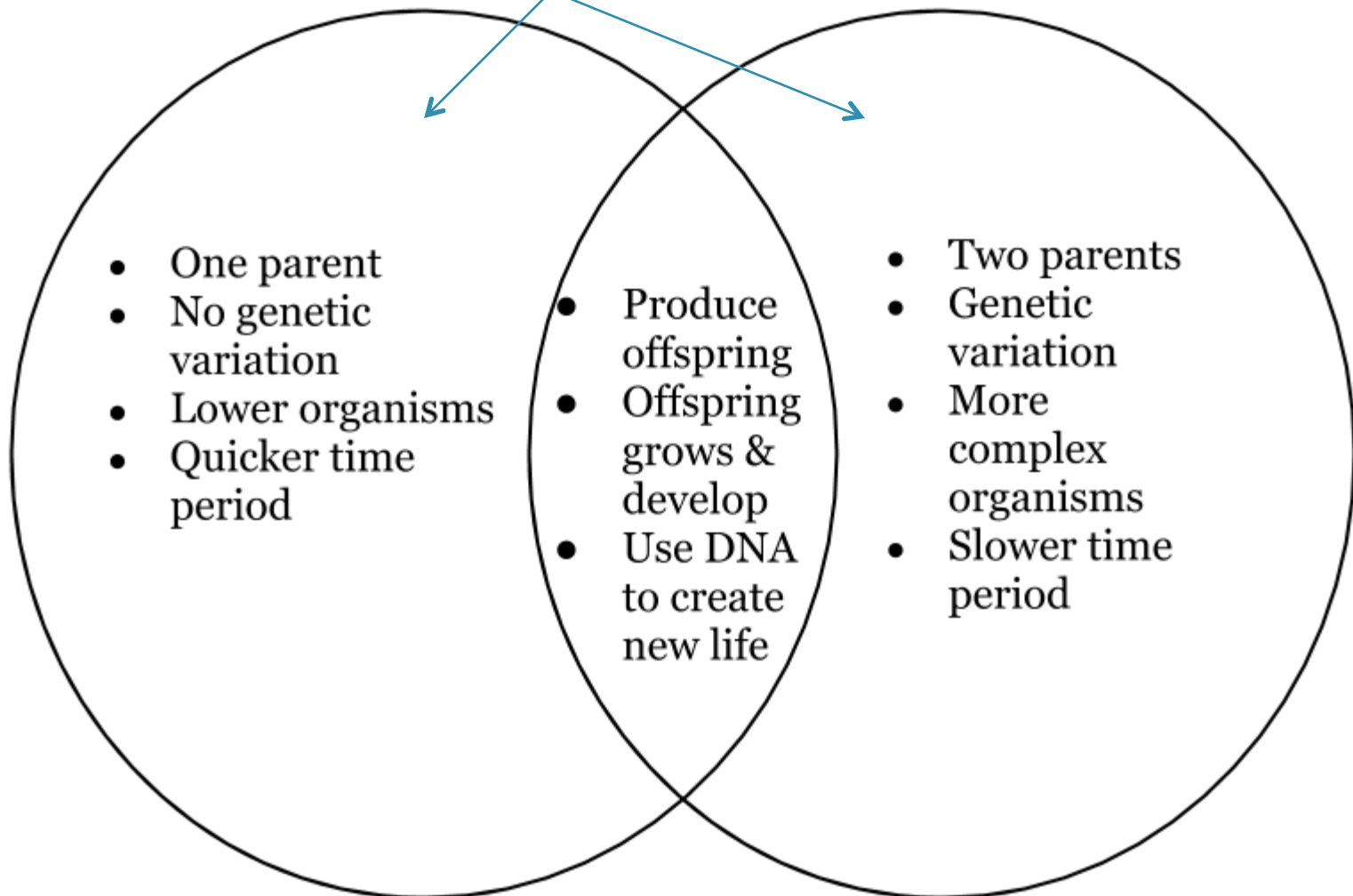
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		U	C	A	G		
1st letter	U	UUU Phe UUC UUA Leu UUG	UCU UCC Ser UCA UCG	UAU Tyr UAC UAA Stop UAG Stop	UGU Cys UGC UGA Stop UGG Trp	U C A G	3rd letter
	C	CUU CUC Leu CUA CUG	CCU CCC Pro CCA CCG	CAU His CAC CAA Gln CAG	CGU CGC Arg CGA CGG	U C A G	
	A	AUU AUC Ile AUA AUG Met	ACU ACC Thr ACA ACG	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G	
	G	GUU GUC Val GUA GUG	GCU GCC Ala GCA GCG	GAU Asp GAC GAA Glu GAG	GGU GGC Gly GGA GGG	U C A G	

3-Cells arise from the division of other cells

- Daughter cells inherit the genes from the mother cells
- Daughter cells inherit cytoplasm and organelles from the mother cells
 - Binary fission - cell division in bacteria
 - Mitosis - the genetic complement of each daughter cell is identical to the other and to the mother cell. This is asexual reproduction
 - Meiosis - the genetic complement of each daughter cell is reduced by half. This is sexual reproduction

The difference between sexual and sexual reproduction as in the following scheme:

Sexual vs. Asexual Reproduction



4-Cells acquire and utilize energy

- Plant cells undergo photosynthesis
 - *convert light energy and CO₂ to chemical energy (ATP and glucose)
- Most cells respire
 - *release energy found in organic compounds
 - *convert organic compounds to CO₂ and O₂
 - *make ATP

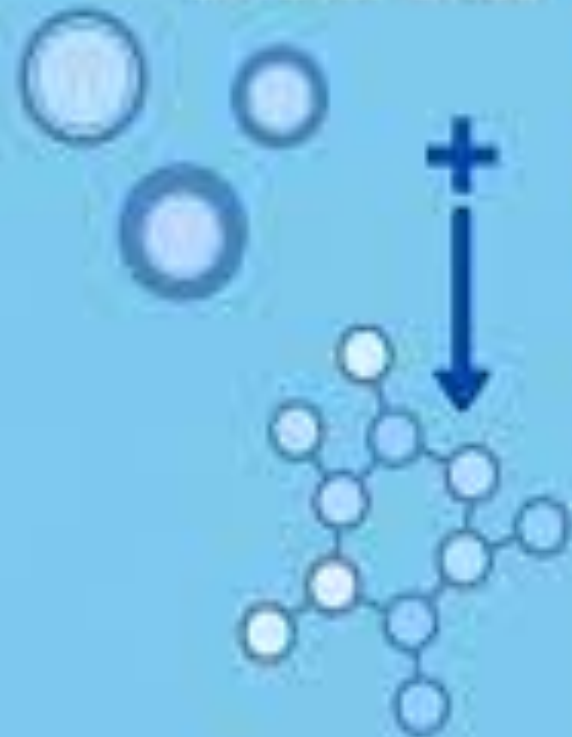
5-Cells can perform a variety of chemical reactions

- convert simple organic molecules into complex molecules (anabolism)
- Breakdown complex molecules to release energy (catabolism)
- Metabolism = all reactions by cells

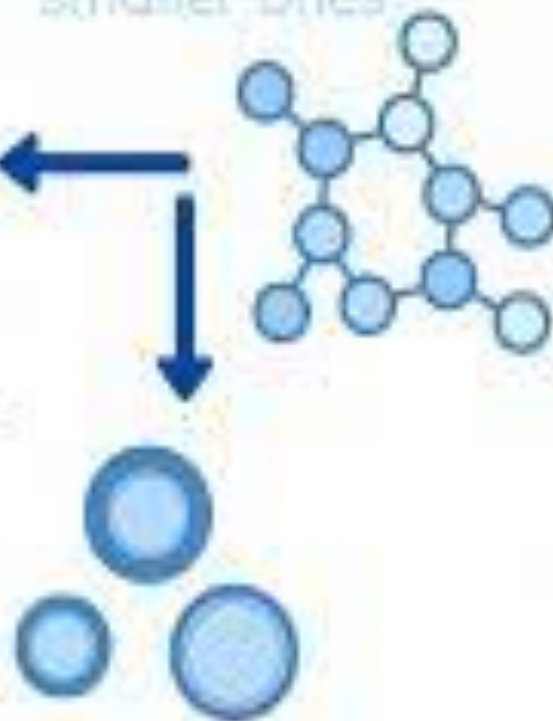


Anabolism and Catabolism

Set of metabolic pathways that synthesize larger molecules from smaller ones



Set of metabolic pathways that break larger molecules into smaller ones



Energy

CATABOLISM
SUPPLIES THE
ENERGY FOR
ANABOLISM

6-Cells can engage in mechanical activities

- Cells can move
- Organelles can move
- Cells can respond to stimuli
 - *chemotaxis - movement towards chemicals
 - *phototaxis - movement towards light
 - *hormone responses
 - *touch responses

7-Cells can regulate activities

- Cells control DNA synthesis and cell division
- Gene regulation - cells make specific proteins only when needed
- Turn on and off metabolic pathways

8-Cells all contain the following structures:

- Plasma membrane - separates the cell from the external environment
- Cytoplasm - fluid-filled cell interior
- Nuclear material - genetic information stored as DNA

- Depending upon their organization, cells are basically of two types
- **Prokaryotic cells and Eukaryotic cells.**
- Prokaryotic cells are primitive type of cells, the cells lacks true nucleus.
- Eukaryotic cells have true nucleus.

Two Different Types of Cells

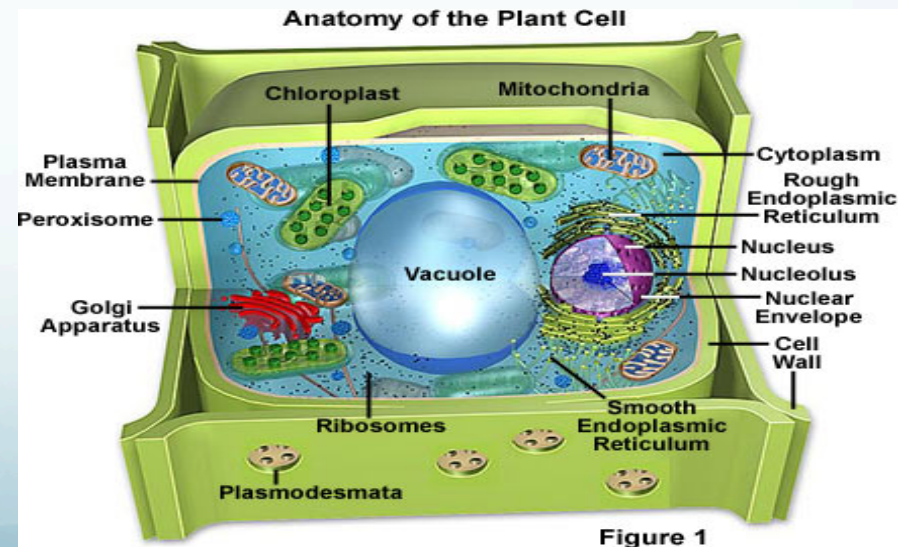
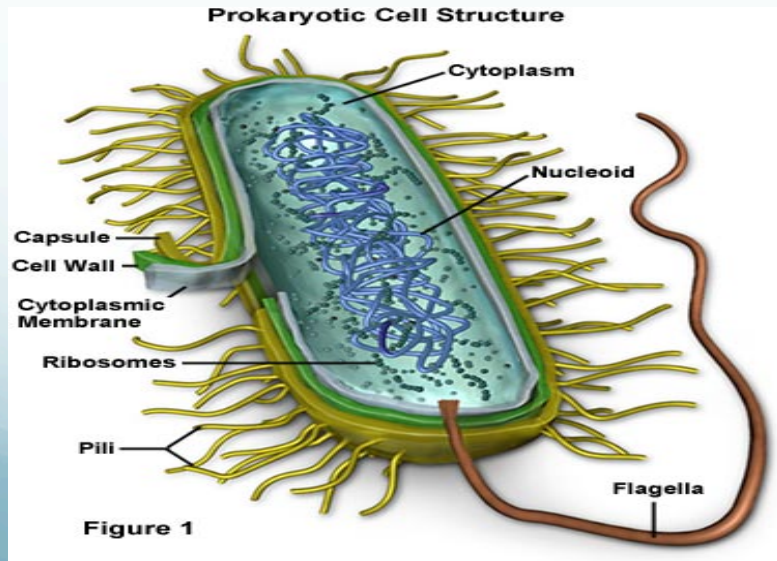
Prokaryotic Cell

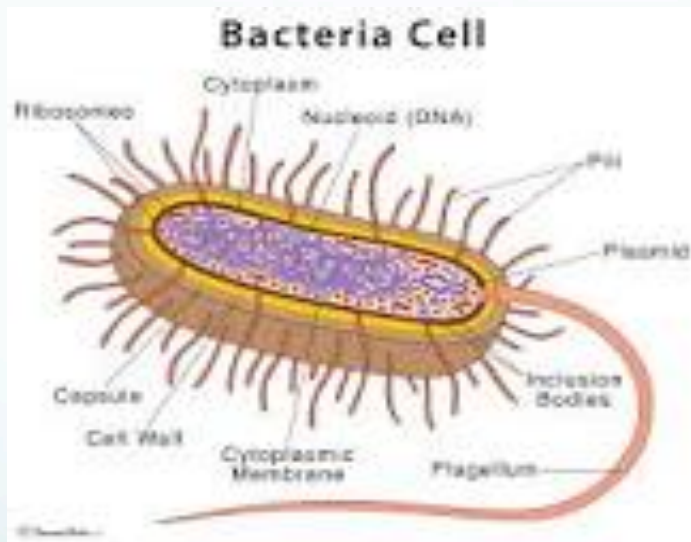
Means “before a nucleus
Ex. : Bacteria, Blue green algae and Archaea.

Eukaryotic Cells

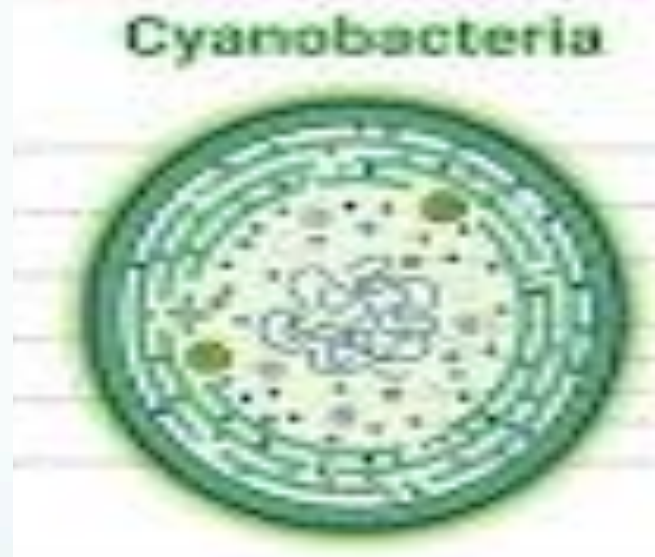
Means “possessing a true nucleus”

- Ex.: Animals, fungi, plants, and protists.





Bacteria

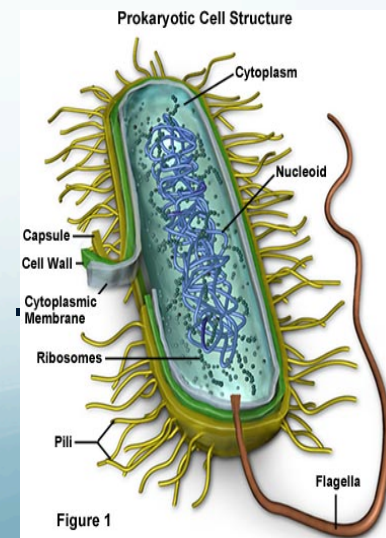


Blue green algae (cyan bacteria)

Prokaryotic Cells

Means “before a nucleus” Ex: *Bacteria*, *Blue green algae*.

- 1- They lack true nuclei enclosed by membrane.
- 2-The genetic material (DNA) is concentrated in a region called the Nucleoid region.
- 3-many prokaryotes have small, circular, double-stranded pieces of ring form DNA called plasmids
- 4- They reproduced only asexually by a mode of cell division called binary-fission.
- 5- They lacks membrane bounded organelles



Prokaryotic Structures

Prokaryotic Cell Structure

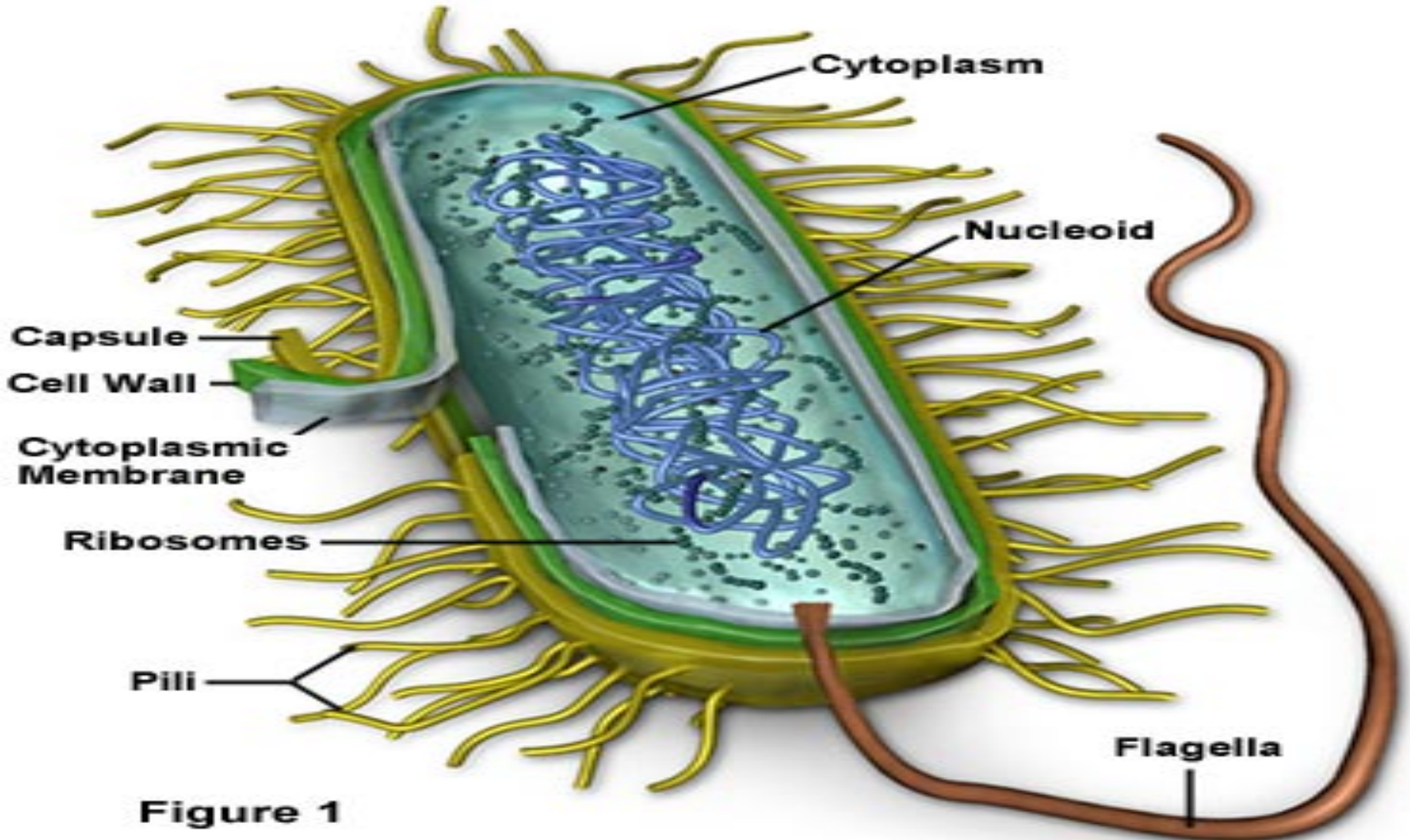


Figure 1

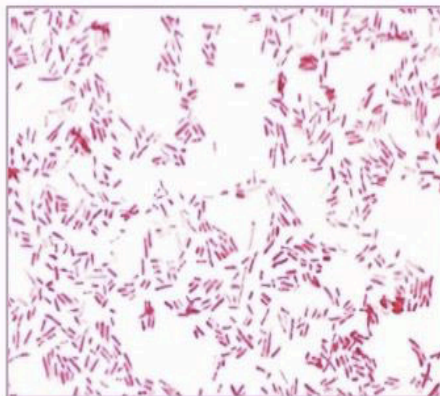
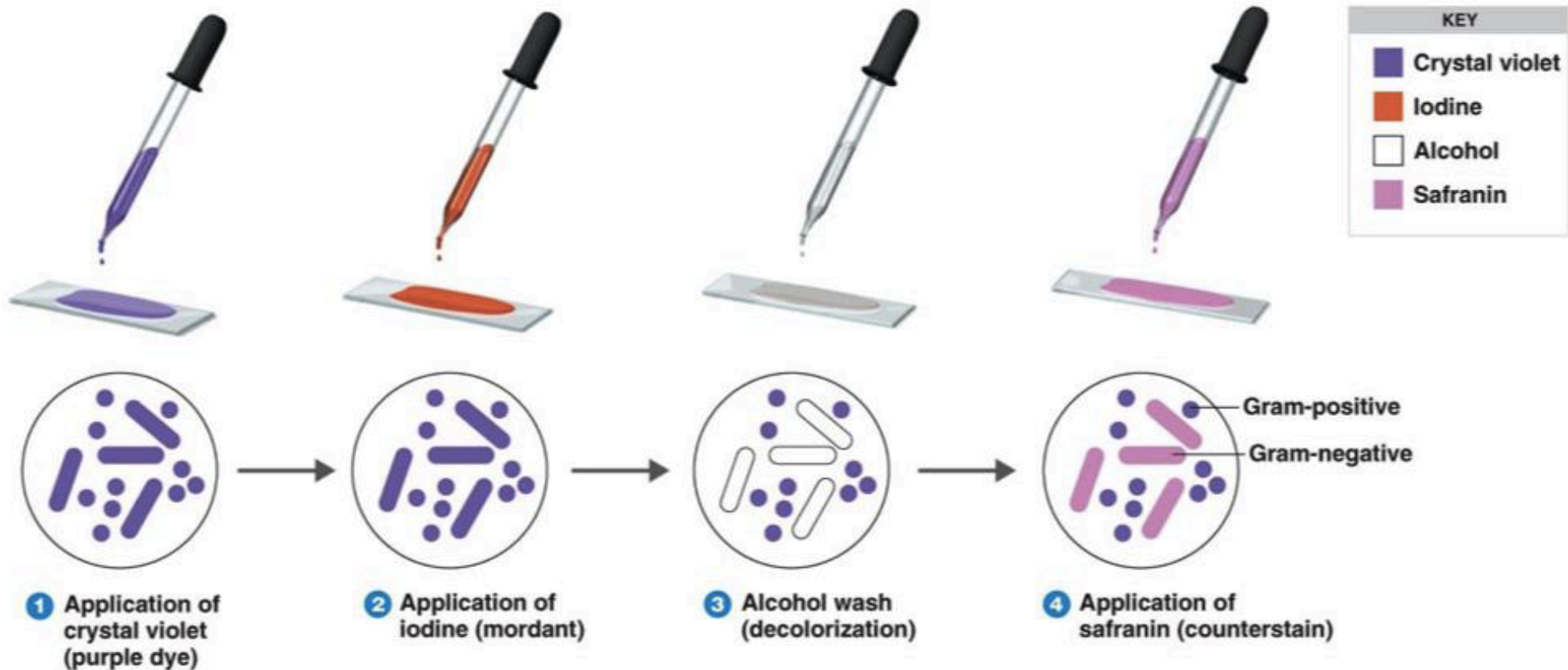
Some structures of prokaryotic cells

Cell Wall

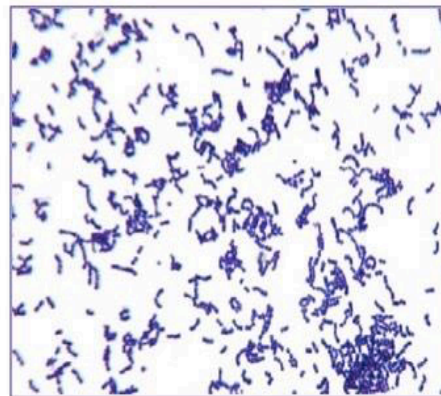
- Supports cell and determines its shape.
- Its an essential structure in bacteria, and its made of chemical components found no where else in nature.
- It may cause symptoms of disease in animals.
- Its the site of action of some of our most important antibiotics.

Structure of the bacterial cell wall:

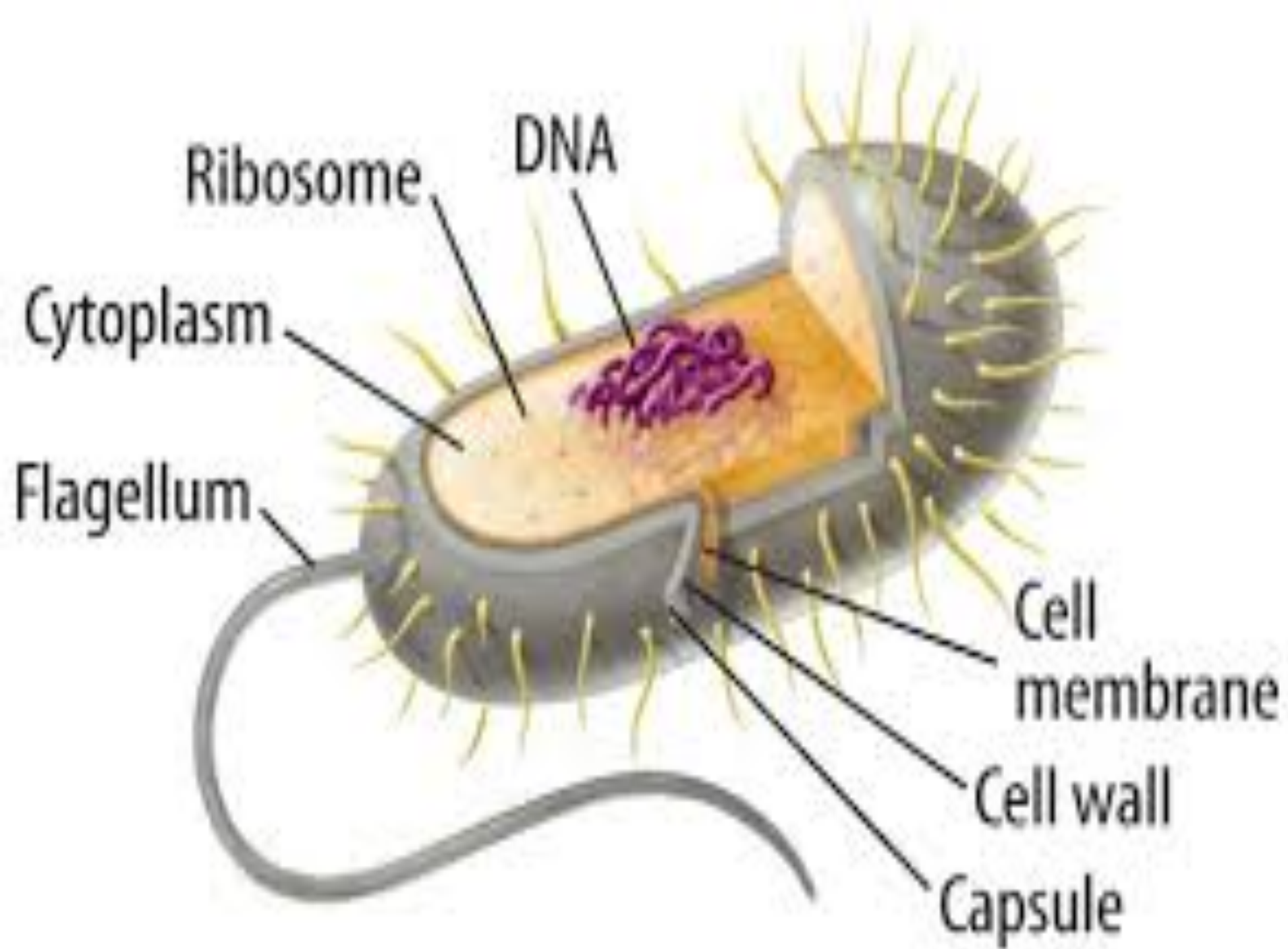
- There are two main types of bacterial cell walls, Gram positive and Gram negative, which are differentiated by their Gram staining characteristics. For both Gram-positive and Gram-negative bacteria, particles of approximately 2 nm can pass through the peptidoglycan.
- Bacterial cell wall provides structural integrity to the cell.
- The bacterial cell wall differs from that of all other organisms by the presence of Peptidoglycan
- Peptidoglycan is responsible for the rigidity of the bacterial cell wall for determination of cell shape.
- Based on the composition of cell wall & staining bacteria are classified into Gram positive and Gram negative



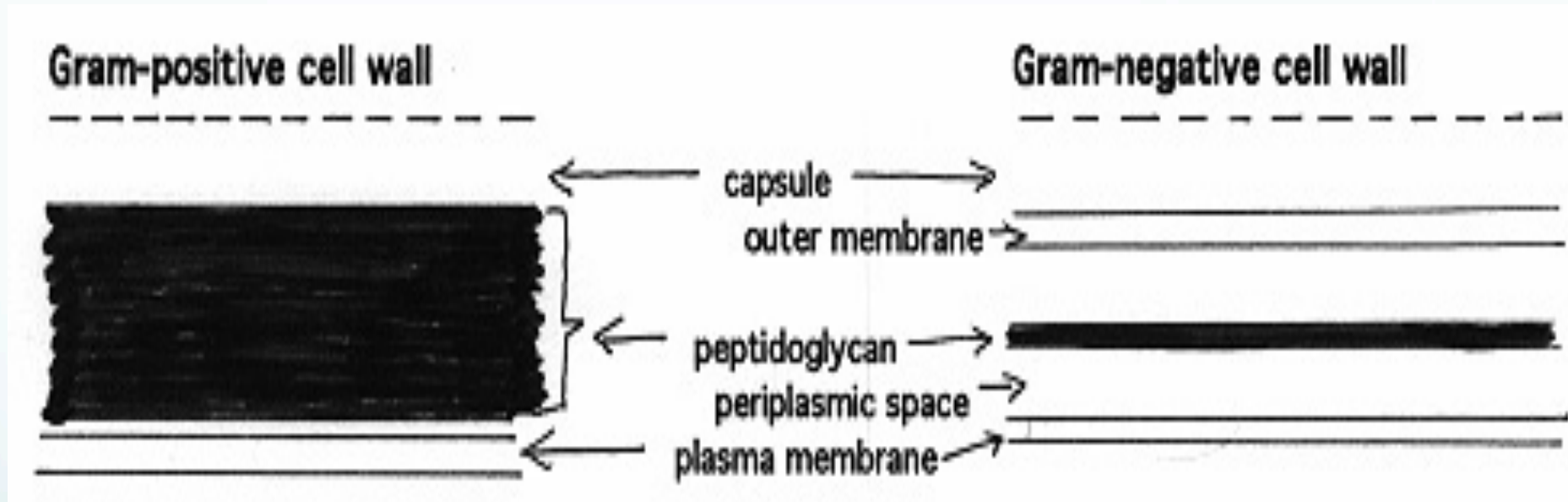
Gram-Negative Bacteria



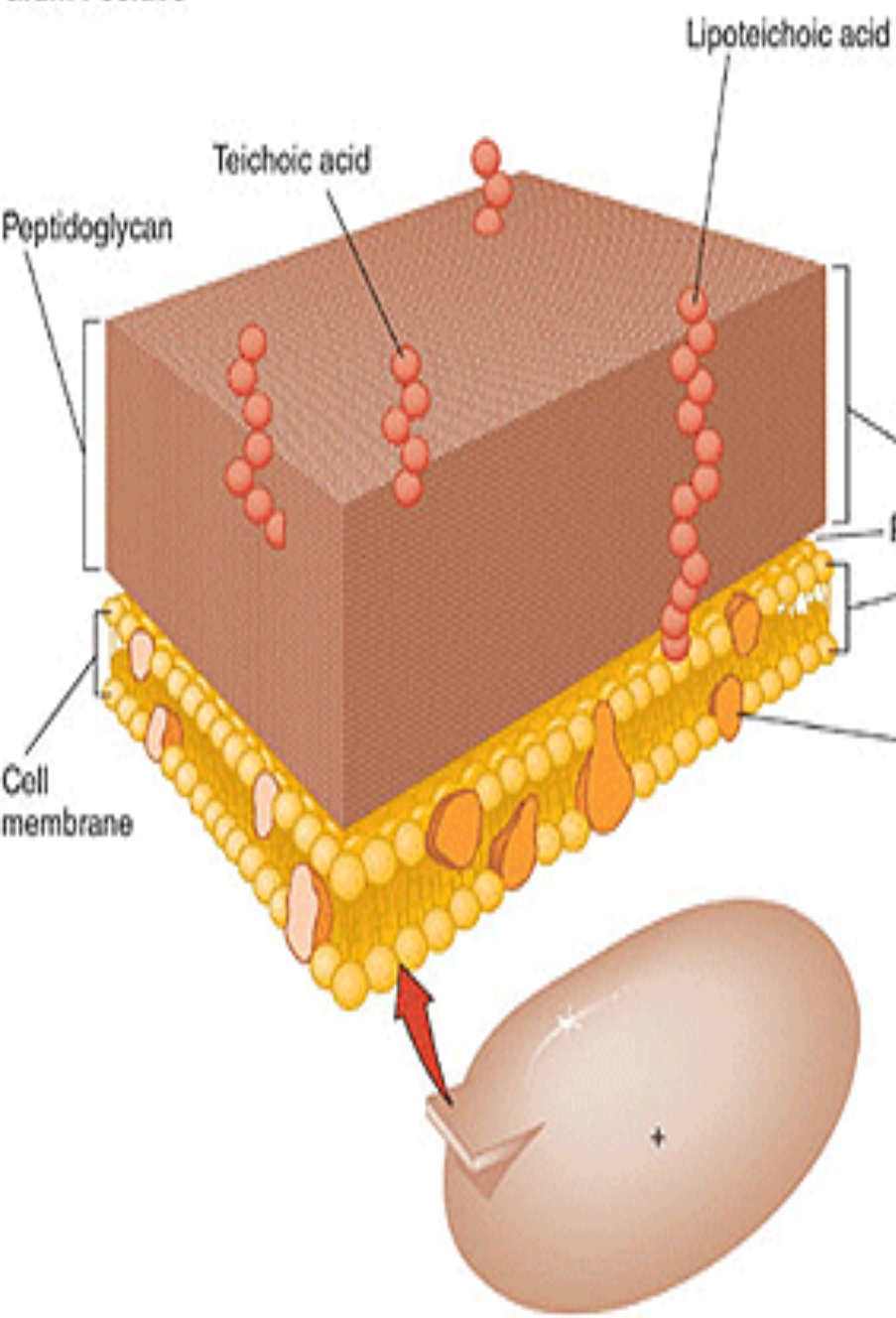
Gram-Positive Bacteria



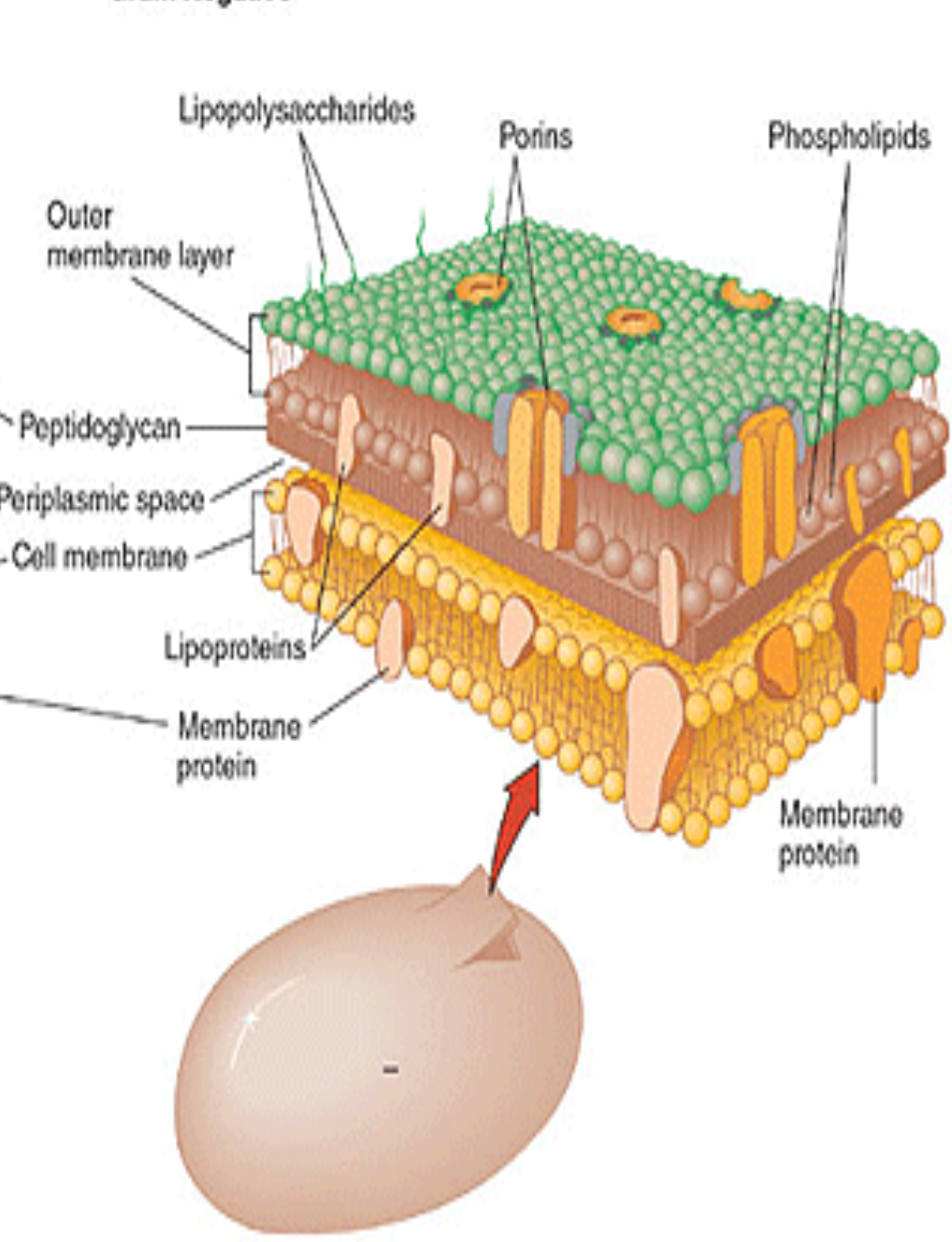
Profile of the bacterial cell envelope (Wall)



Gram Positive



Gram Negative



- Gram-positive bacteria possess a thick cell wall containing many layers of **peptidoglycan**.
- In contrast, Gram-negative bacteria have a relatively thin cell wall consisting of a few layers of peptidoglycan surrounded by a second lipid membrane containing **lipopolysaccharides** and **lipoproteins**.

• Ribosomes

- Ribosomes in prokaryotic cells are consist of two major components: the small ribosomal subunits (RNA), and the large subunits, which join amino acids to form a polypeptide chain (proteins) .
- Prokaryotes have 70 S ribosomes, each consisting of a small(30S) and a large (50S) subunit.
- The main function of ribosomes is protein synthesis



Comparison of features of prokaryotic and eukaryotic cells

	Prokaryotes	Eukaryotes
Typical organisms	Bacteria and Cyanobacter	Protista, Fungi, Animals, Plants
Typical size	~ 1-10µm	~ 10-100µm
Type of nucleus	No real nucleus	Real nucleus with double membrane
DNA	One single circular DNA	Linear Chromatin (DNA molecules associated with histone proteins(chromosomes)).
RNA-/protein-synthesis	RNA is synthesis in the protoplasm	RNA-synthesis inside the nucleus protein synthesis in cytoplasm
Ribosomes	Present with the size (50S+30S)	Present with the size (60S+40S)
Protoplasmatic structure	Very few structures	Highly structured by Endomembranes and a Sytoskeleton
Mitochondria	None	One to several thousand (though some lack mitochondria)
Organization	Usually single cells	Single cells, colonies, higher multicellular organisms with specialized cells
Cell division	Binary fission (simple division)	Mitosis (fission or budding), Meiosis



**THANK
YOU!**